Quantitative Analysis of GFP-KNL-2 Centromeric Signal Dynamics During the First Mitotic Division of a *C. elegans* Embryo Using Fiji/ImageJ

Secondary analysis of Cell Image Library dataset CIL:28778

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Objective

To quantify the change in centre-to-centre distance between GFP-KNL-2 centromeric signals during the first mitosis of a single C. elegans embryo using manual measurements in Fiji/ImageJ.

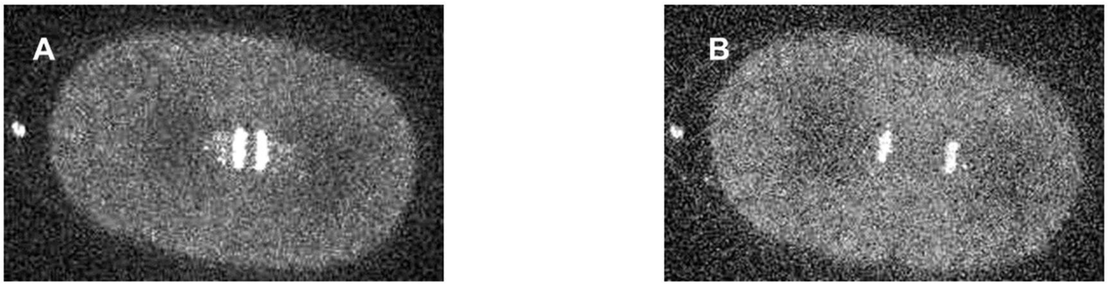
Dataset and Biological Context

This project used publicly available time-lapse microscopy data from Cell Image Library entry CIL:28778. The dataset shows the first mitotic division of a C. elegans embryo expressing GFP-tagged KNL-2, a centromere-associated protein. The original movie contains fluorescence and differential interference contrast (DIC) views; only the fluorescence region was used for quantitative analysis.

Methods

The fluorescence region was isolated from the side-by-side time-lapse movie in Fiji/ImageJ, producing a 252 × 159-pixel fluorescence stack. Identical display settings (0–255) were maintained across all frames. Frames 39–57 were analysed using the straight-line tool to measure the centre-to-centre distance between the two GFP-KNL-2 signals. Distances were recorded in pixels because spatial-calibration metadata were unavailable. Frame numbers were converted to video time using the displayed playback rate of 6 frames/s. Measurements were transferred to Microsoft Excel for calculation and plotting.

GFP-KNL-2 Centromeric Signal Separation



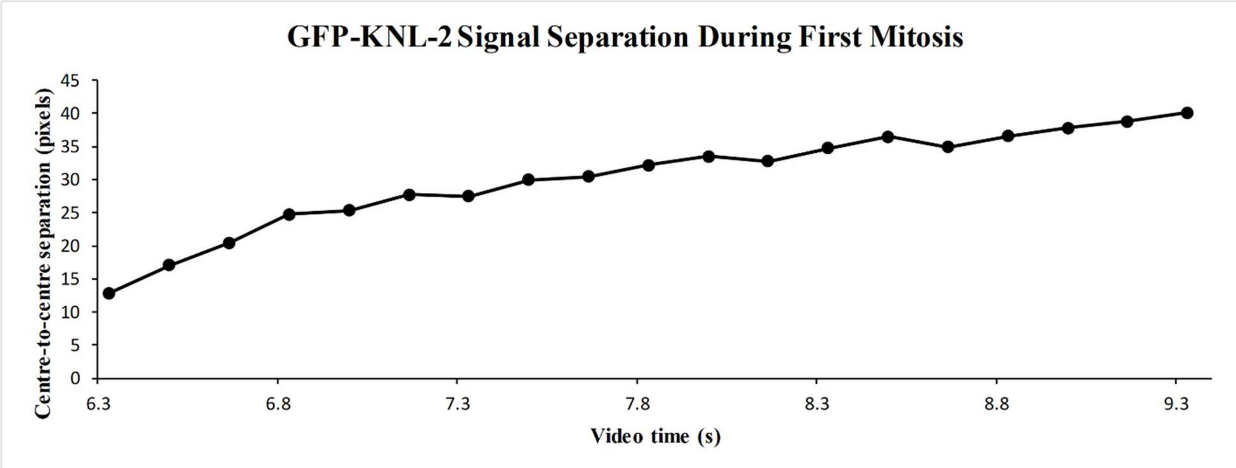
Early separation
Frame 39 | 6.333 s | 12.824 pixels

Later separation
Frame 57 | 9.333 s | 40.106 pixels

Figure 1. Representative fluorescence frames showing increased centre-to-centre separation of GFP-KNL-2 centromeric signals during first mitosis in a C. elegans embryo. (A) Frame 39: video time 6.333 s; separation 12.824 pixels. (B) Frame 57: video time 9.333 s; separation 40.106 pixels. Identical display settings were used.

Results

Across frames 39–57, the measured centre-to-centre distance increased from 12.824 to 40.106 pixels, representing an absolute increase of 27.282 pixels. This corresponds to a 3.127-fold increase, or 212.7%, over 3.0 s of video time. The average apparent separation rate was 9.094 pixels/s. The trajectory showed a generally progressive increase with minor frame-to-frame fluctuations.



Centre-to-centre separation increased from 12.824 to 40.106 pixels over 3.0 s (3.127×; average apparent rate: 9.094 pixels/s).

Figure 2. Centre-to-centre separation of GFP-KNL-2 signals across frames 39–57. Values represent manual measurements from one embryo; time represents displayed video time.

Interpretation

The progressive increase in distance is consistent with separation of the two GFP-KNL-2-positive centromeric structures during first mitosis. Minor decreases between individual frames may reflect biological movement, changes in signal appearance or manual centre-selection variability. Because KNL-2 marks centromeric regions rather than entire chromosomes, these results should be interpreted specifically as GFP-KNL-2 signal separation.

Limitations

- Only one publicly available embryo was analysed.
- Signal centres were selected manually and may contain measurement variability.
- Spatial-calibration metadata were unavailable; therefore, distance is reported in pixels.
- Video time was calculated from the displayed playback rate and may not represent the original acquisition interval.
- No biological replicates or inferential statistical tests were included.

Data Source

Source microscopy material: Cell Image Library entry CIL:28778, licensed under CC BY-NC-SA 3.0. The displayed frames were cropped and annotated for this non-commercial secondary analysis. Available at: https://www.cellimagelibrary.org/images/28778 (accessed 3 September 2026).

Schindelin, J., et al. (2012). Fiji: an open-source platform for biological-image analysis. Nature Methods, 9, 676–682. https://doi.org/10.1038/nmeth.2019